

SUOMUSSALMI PESIO, Finland

WMO: 028890

Lat: 64.9314N

Lon: 28.7467E

Elev: 225

StdP: 98.65

Time Zone: 2.00 (EUE)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.6	-27.4	-33.4	0.2	-30.3	-29.7	0.2	-27.2	6.4	-5.7	5.9	-6.5	0.9	300	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	25.3	17.4	23.2	16.2	21.3	15.4	19.0	23.1	17.7	21.4	16.4	19.9	2.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.6	13.0	20.6	16.2	11.8	19.3	15.0	10.9	18.0	54.9	22.9	50.6	21.5	46.9	19.8	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.1	5.4	4.8	-33.5	27.3	3.5	2.4	-36.0	29.0	-38.1	30.4	-40.0	31.8	-42.5	33.5
			-33.4	20.3	3.4	2.0	-35.9	21.8	-37.9	22.9	-39.8	24.1	-42.3	25.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.4	-11.5	-10.7	-6.6	0.0	6.8	11.4	15.3	12.9	8.0	1.4	-3.7	-7.7
	DBStd	10.49	8.07	7.45	5.87	4.25	4.59	3.68	3.38	3.28	3.36	4.11	5.26	7.13
	HDD10.0	3526	667	580	514	301	121	26	3	11	77	267	410	550
	HDD18.3	6208	926	813	772	551	358	211	105	170	309	525	660	808
	CDD10.0	374	0	0	0	0	21	67	167	101	17	1	0	0
	CDD18.3	15	0	0	0	0	1	2	11	2	0	0	0	0
	CDH23.3	164	0	0	0	0	9	32	100	23	0	0	0	0
	CDH26.7	18	0	0	0	0	0	4	12	2	0	0	0	0
Wind	WSAvg	2.3	2.0	2.1	2.5	2.4	2.6	2.4	2.1	2.0	2.2	2.4	2.3	2.1
Precipitation	PrecAvg	667	45	38	33	32	55	62	89	80	66	60	53	53
	PrecMax	931	88	70	57	55	95	137	148	151	118	150	107	113
	PrecMin	490	7	8	8	11	18	15	22	26	29	10	21	8
	PrecStd	101	20	18	13	15	22	27	31	33	27	33	23	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.4	3.6	7.7	14.9	24.7	27.1	28.2	26.1	19.4	11.5	7.7	3.3
		MCWB	1.8	2.3	3.2	7.9	13.3	16.8	19.3	18.2	14.8	10.0	7.3	2.9
	2%	DB	0.6	1.4	4.1	11.0	21.8	23.4	26.1	23.2	16.8	9.5	5.6	2.0
		MCWB	0.4	0.8	1.5	5.8	13.3	15.8	18.5	17.2	13.1	8.7	5.2	1.6
	5%	DB	-0.1	0.1	2.2	8.8	18.3	20.9	24.0	21.0	14.9	8.0	3.6	1.0
		MCWB	-0.3	-0.1	0.4	4.0	11.2	13.8	17.2	15.9	12.2	7.4	3.3	0.7
	10%	DB	-1.3	-1.1	0.9	6.8	15.3	18.8	22.1	19.2	13.4	6.9	2.1	0.1
		MCWB	-1.5	-1.4	-0.2	3.2	9.2	12.9	16.3	15.2	11.5	6.3	1.9	0.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.9	2.5	3.5	8.7	15.1	18.8	21.2	20.1	15.7	10.5	7.5	3.2
		MCDB	2.6	3.4	6.8	12.9	21.6	24.0	25.0	24.0	18.3	11.2	7.7	3.3
	2%	WB	0.3	0.9	1.8	6.4	13.6	16.8	19.8	18.1	13.9	8.8	5.2	1.7
		MCDB	0.4	1.4	3.3	10.3	19.9	21.4	23.8	21.4	15.9	9.3	5.5	1.9
	5%	WB	-0.2	-0.1	0.9	4.7	11.8	15.2	18.5	16.8	12.7	7.5	3.4	0.7
		MCDB	-0.2	0.1	1.9	7.8	17.7	19.4	22.4	19.9	14.3	7.9	3.6	0.9
	10%	WB	-1.5	-1.3	0.1	3.4	10.0	13.8	17.2	15.6	11.7	6.4	1.9	0.1
		MCDB	-1.3	-1.1	0.8	6.1	14.3	17.5	20.8	18.4	13.0	6.9	2.1	0.2
Mean Daily Temperature Range	5% DB	MDBR	6.5	7.7	9.8	9.6	10.1	9.6	9.5	8.8	7.1	4.5	4.4	5.8
		MCDBR	5.1	5.3	9.5	13.3	15.9	14.0	12.7	12.3	9.8	5.6	4.1	4.4
		MCWBR	5.2	5.1	7.4	8.2	8.2	6.8	5.8	6.7	6.4	4.7	4.1	4.3
	5% WB	MCDBR	5.0	5.2	7.2	11.6	14.1	11.4	11.3	10.8	8.6	5.2	4.1	3.9
		MCWBR	5.1	5.0	5.9	7.4	7.7	6.1	5.9	6.4	6.1	4.6	4.1	3.9
Clear-Sky Solar Irradiance	taub	0.198	0.268	0.298	0.332	0.339	0.339	0.361	0.340	0.317	0.293	0.350	0.198	
	taud	1.931	2.114	2.163	2.257	2.464	2.486	2.444	2.481	2.532	2.386	2.658	1.958	
	Ebn at Noon	315	615	771	835	871	877	842	828	774	606	299	89	
	Edh at Noon	24	59	89	104	94	94	95	83	64	46	21	8	
All-Sky Solar Radiation	RadAvg	0.12	0.63	2.05	3.75	4.65	5.13	4.82	3.57	1.95	0.69	0.16	0.04	
	RadStd	0.01	0.13	0.19	0.38	0.48	0.36	0.49	0.35	0.21	0.10	0.03	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	+0.57	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page